

MoE-LLM: A Hybrid Ensemble Framework Combining Mixture of Experts and LLM-Agent for Traffic Prediction

ANONYMOUS AUTHOR(S)

Accurate mobile traffic prediction is essential for efficient network resource allocation and stable service delivery. However, existing forecasting methods often rely on a single architecture to model heterogeneous spatio-temporal patterns, while existing Mixture-of-Experts (MoE) approaches mainly coordinate experts through learned numerical gating, limiting their ability to reason about expert suitability and adapt to evolving traffic conditions. To address these limitations, we propose MoE-LLM, an LLM-based agent framework for coordinating multiple complementary experts in mobile traffic prediction. Specifically, temporal experts capture trend, seasonality, and residual dynamics, while spatial experts model geographic, functional, and traffic-pattern relationships from complementary perspectives. An LLM-based coordination agent then reasons over expert predictions, expert profiles, and spatio-temporal context to generate the final prediction. Moreover, a feedback mechanism evaluates recent expert performance and dynamically updates expert reliance, enabling adaptive coordination as traffic patterns evolve. The agent also provides interpretable reasoning for expert prioritization and reliance adjustment. Extensive experiments on real-world traffic datasets demonstrate that our proposed model surpasses state-of-the-art methods by over 5% in mobile traffic prediction and is highly interpretable. The code is available at <https://anonymous.4open.science/r/MoELLM-ACD4>.

CCS Concepts: • **Information systems** → **Spatial-temporal systems**; • **Networks** → *Mobile networks*; • **Computing methodologies** → *Machine learning algorithms*.

Additional Key Words and Phrases: Traffic Prediction; Mixture of Experts; Large Language Model

1 Introduction

The rapid growth of mobile services and increasing pressure on cellular network resources has caused traffic congestion and transmission delays. Accurate mobile traffic prediction is therefore essential for efficient resource allocation and stable service delivery [4, 23]. Existing methods for traffic prediction have evolved from sequential modeling [5, 17] to context-aware spatio-temporal models [8, 9]. Most existing methods rely on a single forecasting architecture to model diverse spatio-temporal patterns, making it difficult to capture their heterogeneous characteristics with dedicated modeling mechanisms. To address this limitation, Mixture-of-Experts (MoE) frameworks, such as MoGERNN [29] and ST-ExpertNet [19], employ multiple specialized experts to capture distinct traffic patterns and aggregating their complementary predictions. Nevertheless, existing MoE-based traffic predictors typically coordinate experts through learned gating or routing functions. Although these mechanisms can assign different weights to experts for different inputs, the coordination rule itself remains fixed after training and is mainly driven by numerical features. As traffic conditions evolve, they cannot explicitly reassess expert reliability based on recent prediction outcomes or revise their reliance on experts that become less suitable. This limits their ability to continuously adapt expert coordination under non-stationary traffic dynamics.

Large Language Models (LLMs) have demonstrated strong capabilities in semantic reasoning and adaptive decision-making, making them promising for coordinating multiple specialized experts. By jointly reasoning over expert predictions, spatio-temporal context, and historical reliability, an LLM-based agent can assess the relative suitability of different experts under varying traffic conditions. Moreover, feedback from recent prediction errors can be incorporated to continuously adjust expert reliance, enabling adaptive coordination as traffic dynamics evolve. For instance, TrafficGPT [28] demonstrates the capability of foundation models to understand and reason over heterogeneous traffic information across multiple tasks, while FLAIRR-TS [6] shows that LLM agents can iteratively refine time-series forecasts through feedback-driven test-time optimization.

These studies demonstrate the potential of LLM-based agents to combine contextual reasoning with adaptive refinement for traffic prediction.

In this paper, we aim to design an LLM-based agent framework to coordinate specialized experts for mobile traffic prediction. Achieving this goal requires addressing the following challenges. (1) **How to design effective experts that capture spatio-temporal dependencies?** Mobile traffic exhibits heterogeneous temporal dynamics and complex spatial correlations, which cannot be sufficiently characterized from a single modeling perspective. It is therefore essential to design complementary experts that specialize in distinct temporal patterns and spatial dependencies. (2) **How to enable the agent to perform semantic reasoning over multiple expert predictions?** Different experts may produce inconsistent or even conflicting predictions due to their distinct modeling focuses. Rather than simply aggregating these outputs, the agent needs to reason over expert predictions together with the current spatio-temporal context and determine how their complementary information should be synthesized into a coherent final prediction. (3) **How to enable the agent to adaptively adjust its reliance on experts?** The reliability of each expert may vary across different traffic conditions. Therefore, the agent needs to continuously evaluate expert performance and dynamically adjust its reliance on each expert as traffic patterns evolve.

To overcome these challenges, we propose MoE-LLM, an agent-based framework that models spatio-temporal dependencies through complementary specialized experts and leverages an LLM agent for semantic coordination and adaptive expert reliance. First, a mixture-of-experts architecture captures heterogeneous spatio-temporal dependencies from complementary perspectives, where temporal experts characterize trend, seasonality, and residual dynamics across different temporal patterns, while spatial experts model geographic proximity, functional similarity, and traffic-pattern correlation. Second, an LLM-based agent semantically coordinates multiple experts by reasoning over their predictions, specialized roles, and the current spatio-temporal context, enabling context-aware integration into a unified final prediction. Third, an adaptive coordination mechanism enables the agent to continuously assess expert reliability based on recent prediction errors and dynamically adjust reliance on different experts as traffic conditions evolve.

Our main contributions can be summarized as follows:

- We investigate the use of LLM-based agents for coordinating mixture-of-experts in mobile traffic prediction. We explore how semantic reasoning and adaptive decision-making can enable effective collaboration among specialized experts under heterogeneous and evolving traffic conditions.
- We propose MoE-LLM, which integrates complementary temporal and spatial experts with an LLM-based coordination agent. The agent jointly reasons over expert predictions and spatio-temporal context, while a feedback mechanism continuously updates expert reliance to adapt their contributions over time.
- Extensive experiments on two real-world mobile traffic datasets demonstrate that MoE-LLM consistently outperforms state-of-the-art baselines, reducing both MAE and RMSE by over 5%. Further analyses verify the effectiveness of expert specialization, LLM-based coordination, and adaptive expert reliance.

2 Related Work

MoE for Time Series Prediction. The MoE paradigm dynamically routes inputs to specialized sub-models. Early time series approaches decompose data into trend and seasonal components [15], while sparse MoE structures [24] improve scalability. MoE has been widely adopted in NLP [2, 3] and adapted to time series for capturing multi-scale patterns with shallow or frequency-based gating [14]. In contrast, our framework uses an LLM to dynamically integrate multiple forecasting experts.

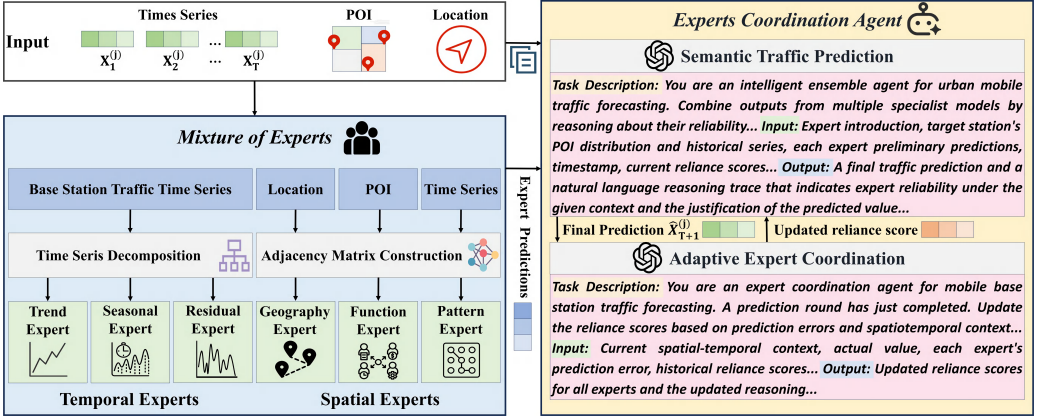


Fig. 1. Overall framework of the proposed MoE-LLM. The framework consists of temporal experts and spatial experts, and an LLM-based experts coordination agent. The agent semantically integrates initial experts' predictions, adaptively coordinating experts via error analysis and dynamic reliability scoring, and outputs the final prediction.

LLM for Time Series Prediction. LLMs have shown strong reasoning capabilities and are increasingly used for time series forecasting. Recent studies adapt LLMs via cross-modal alignment to map time series into LLM-compatible spaces [11, 30] or prompt-based methods to inject temporal context [25, 27]. However, existing approaches typically rely on a single model and cannot dynamically coordinate multiple experts. To address this, we propose an LLM-based expert coordination agent that adaptively integrates multiple forecasting experts through context-aware reasoning.

3 Problem Definition

In this study, we focus on the task of traffic prediction. Given historical traffic observations and regional environmental context, the goal is to predict future traffic values. Mathematically, given the historical time series $X^{(j)} = (x_1^{(j)}, \dots, x_T^{(j)}) \in \mathbb{R}^T$ of T steps and environmental context $C^{(j)}$ for each base station $j \in \{1, \dots, V\}$, we aim to design a model f_Θ that predicts the traffic value $\hat{x}_{T+1}^{(j)}$ at the next time step:

$$\hat{x}_{T+1}^{(j)} = f_\Theta(X^{(j)}, C^{(j)}). \quad (1)$$

4 Method

The overall framework of the proposed MoE-LLM is illustrated in Figure 1. The MoE-LLM consists of spatio-temporal experts for generating preliminary predictions from multiple perspectives, and an LLM-based coordination agent that leverages an LLM to reason the final prediction and adaptively adjusts expert reliance based on their prediction performance.

4.1 Temporal Experts

To capture heterogeneous temporal dynamics, we decompose the traffic time series into distinct components and employ specialized experts to model corresponding patterns. We first use fourier decomposition to separate the traffic time series into trend, seasonal, and residual components, characterizing traffic dynamics across different temporal scales and functional roles. Formally, given the traffic time series $x_t^{(j)}$ of base station j at time step t , the decomposition process is:

$$x_t^{(j)} = x_{\text{trend},t}^{(j)} + x_{\text{seasonal},t}^{(j)} + x_{\text{residual},t}^{(j)}, \quad (2)$$

where $x_{\text{trend},t}^{(j)}$, $x_{\text{seasonal},t}^{(j)}$, and $x_{\text{residual},t}^{(j)}$ represent the trend, seasonal, and residual components, respectively. These components characterize the traffic dynamics from different temporal aspects and functional roles. Specifically, The trend component captures long-term variation, the seasonal component captures periodic patterns, and the residual component accounts for irregular variations caused by unexpected events or noise. This decomposition enables separate learning of distinct temporal dynamics for more accurate and interpretable forecasting. We then employ dedicated temporal experts to model distinct temporal dynamics. Specifically, the trend expert adopts a linear regression model to capture smooth long-term trends. The seasonal expert combines Fourier transformation with an LSTM [5] to model periodic patterns and sequential dependencies. The residual expert employs a diffusion-based model to capture irregular and stochastic variations that are difficult to model deterministically.

4.2 Spatial Experts

We model spatial dependencies from three complementary perspectives. For each perspective, we construct a dedicated graph and employ a spatio-temporal graph neural network to capture the corresponding spatial interactions among base stations. These experts characterize local geographic influence, functional relatedness, and long-range traffic-pattern similarity, providing complementary spatial information for traffic prediction.

4.2.1 Geography Expert. Constructing spatial relationships based on geographical distance has been widely adopted in traffic prediction [1], as adjacent base stations tend to exhibit similar traffic fluctuations due to user mobility continuity and signal propagation. We construct an adjacency matrix based on the Haversine distances between base stations, capturing the direct impact of spatial proximity on traffic variations. The adjacency weight between two nodes is computed using an exponential decay function over their squared distance, formulated as:

$$A_{ij}^{(D)} = \exp\left(-\frac{d_{ij}^2}{\sigma^2}\right), \quad \bar{A}_{ij}^{(D)} = \begin{cases} A_{ij}^{(D)}, & \text{if } A_{ij}^{(D)} \geq \tau, \\ 0, & \text{otherwise.} \end{cases}, \quad (3)$$

where d_{ij} denotes the Haversine distance between base station i and j , and σ controls the decay rate. A threshold τ is introduced to avoid the computational overhead caused by dense graphs.

4.2.2 Function Expert. Regions with similar urban functions tend to exhibit similar traffic patterns, as human mobility and service demands are largely shaped by the socioeconomic activities associated with different land uses [31]. Such functional characteristics can be effectively captured by Point-of-Interest (POI) compositions [20], which reflect the types and density of services available in each region. Following this intuition, we construct a POI-based representation for each base station. Specifically, each POI record includes a categorical label. To systematically characterize urban functions, we group these fine-grained labels into 14 macro-categories. Based on these categories, we construct a POI distribution vector P_i for each base station. For a given base station i , we define its spatial neighborhood as a circular region of radius $r = 1.0$ km centered at its geographic coordinates. Within this neighborhood, we count the number of POIs belonging to each macro-category $c \in \{1, \dots, 14\}$, forming a raw count vector $\tilde{P}_i \in \mathbb{R}^{14}$, where $\tilde{P}_i[c]$ is the count of POIs of category c . To ensure that the similarity reflects functional composition rather than absolute POI density, we apply L2 normalization: $P_i = \frac{\tilde{P}_i}{\|\tilde{P}_i\|_2 + \epsilon}$, where $\epsilon = 10^{-8}$ prevents division by zero. For base stations with no POI in their neighborhood, we set $P_i = \mathbf{0}$. The functional similarity between two base stations i and j is then defined as the cosine similarity between their normalized

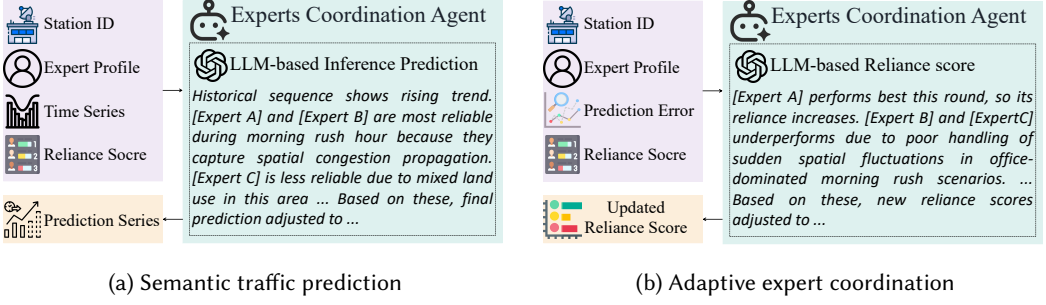


Fig. 2. Semantic reasoning and adaptive coordination of the LLM-based experts coordination agent. The prediction phase integrates expert predictions, expert profiles, spatio-temporal context, and historical reliance scores to generate the final forecast, while the coordination phase analyzes recent prediction errors to dynamically update expert reliance for subsequent predictions.

POI vectors:

$$A_{ij}^{(F)} = \frac{P_i \cdot P_j}{\|P_i\| \|P_j\|}, \quad \tilde{A}_{ij}^{(F)} = \begin{cases} A_{ij}^{(F)}, & \text{if } A_{ij}^{(F)} \geq \tau, \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

This ensures that base stations with similar functional compositions receive high adjacency weights, enabling the model to capture semantic spatial dependencies beyond physical proximity. A threshold τ is also introduced to extract a sparse set of reliable functional neighbors.

4.2.3 Pattern Expert. Different base stations may exhibit similar rush-hour patterns, indicating potential traffic pattern correlations even when they are geographically distant or functionally dissimilar [12, 13]. To capture such dynamics-driven similarities, we construct a pattern expert that uses Pearson correlation to define adjacency, assigning higher connectivity to nodes with similar temporal evolution patterns. Specifically, we compute the Pearson correlation coefficient between the historical traffic sequences of each pair of base stations i and j as:

$$A_{ij}^{(P)} = \frac{\sum_t (X_t^{(i)} - \bar{X}^{(i)})(X_t^{(j)} - \bar{X}^{(j)})}{\sqrt{\sum_t (X_t^{(i)} - \bar{X}^{(i)})^2} \sqrt{\sum_t (X_t^{(j)} - \bar{X}^{(j)})^2}}, \quad \tilde{A}_{ij}^{(P)} = \begin{cases} A_{ij}^{(P)}, & \text{if } A_{ij}^{(P)} \geq \tau, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

where $X_t^{(i)}$ and $X_t^{(j)}$ denote the traffic flow of base station i and j at time step t , respectively, and $\bar{X}^{(i)} = \frac{1}{T} \sum_{t=1}^T X_t^{(i)}$ is the temporal mean of base station i over the historical window. This formulation yields values in $[-1, 1]$, where positive values indicate positively correlated traffic patterns.

4.3 Experts Coordination Agent

Temporal experts and spatial experts have been designed previously to capture the spatiotemporal dependence of base station traffic from different perspectives. However, how to effectively integrate the prediction results of these experts to form a unified final prediction remains a key challenge. Different experts perform differently under varying traffic regimes. To address this, We propose an expert coordination agent that integrates temporal and spatial context from experts and performs semantic reasoning to produce a final traffic prediction. The agent automatically orchestrates experts by evaluating their performance and adaptively adjusting their contributions in subsequent predictions.

4.3.1 Semantic Traffic Prediction. Different experts excel under varying traffic regimes, and their relative strengths shift with the current traffic context. A simple weighted average cannot capture such dynamics. We therefore adopt a context-aware selection strategy, where expert contributions are determined by the specific scenario at hand. To operationalize this context-aware selection, we design a structured context interface that is automatically instantiated for each prediction. This interface explicitly organizes the task specification $\mathcal{T}$, the experts' profiles $\mathcal{E}$, the spatio-temporal context $C^{(j)}$ of the target base station j , the preliminary predictions $\hat{x}_{n,T+1}^{(j)}$ from N experts ($n \in \{1, 2, \dots, N\}$), and their corresponding reliance scores $w_{n,T}^{(j)}$. Specifically, $\mathcal{T}$ defines the prediction task, $\mathcal{E}$ describes the design rationale and specialization of each expert, and $C^{(j)}$ encodes the temporal and spatial context of the target base station. This structured formulation enables the LLM to reason holistically over both the numerical forecasts and their underlying semantic context. The final prediction $\hat{x}_{T+1}^{(j)}$ is generated through the LLM-driven reasoning process, which can be formally represented as:

$$\hat{x}_{T+1}^{(j)} = f_{\text{LLM}} \left(\mathcal{P} \left(\mathcal{T}, \mathcal{E}, C^{(j)}, \{\hat{x}_{n,T+1}^{(j)}\}_{n=1}^N, \{w_{n,T}^{(j)}\}_{n=1}^N \right) \right), \quad (6)$$

where $f_{\text{LLM}}(\cdot)$ denotes the generation function of the language model, and $\mathcal{P}(\cdot)$ represents the prompt construction function that converts the numerical predictions and contextual metadata into natural language instructions. As illustrated in Figure 2(a). The LLM is instructed to analyze the current context and expert reliability, then output a object containing the final prediction and its reasoning trace.

4.3.2 Adaptive Experts Coordination. After each prediction, the experts coordination agent updates each expert's reliance score based on its recent predictive performance, so that subsequent predictions can place greater emphasis on more reliable experts and adapt to evolving traffic conditions. Specifically, once the ground truth traffic volume $x_{T+1}^{(j)}$ becomes available, the framework enters the coordination phase, where the n -th expert's prediction error $e_{n,T+1}^{(j)}$ is evaluated as the absolute difference between the ground truth and the preliminary prediction: $e_{n,T+1}^{(j)} = \left(x_{T+1}^{(j)} - \hat{x}_{n,T+1}^{(j)} \right)^2$. Based on this evaluation and the current spatio-temporal context $C^{(j)}$, the reliance score $w_{n,T+1}^{(j)}$ is generated by the LLM through semantic analysis and reasoning. The updated reliance generation process is given as:

$$\{w_{n,T+1}^{(j)}\}_{n=1}^N = f_{\text{LLM}} \left(Q \left(\mathcal{T}', C^{(j)}, \{e_{n,T+1}^{(j)}\}_{n=1}^N, \{w_{n,T}^{(j)}\}_{n=1}^N \right) \right), \quad (7)$$

where $Q(\cdot)$ denotes the prompt construction function tailored for coordination update, $\mathcal{T}'$ denotes the task specification, and $e_{n,T+1}^{(j)}$ denotes the prediction error. This guides the LLM to reassess each expert's reliability based on recent errors and historical reliance distributions. These updated scores feed back into the context interface for the next prediction, forming an automated feedback loop. As illustrated in Figure 2(b). The LLM is instructed to analyze the current context and each expert's prediction error, then output updated reliance scores for the next prediction round.

5 Experiments

5.1 Datasets.

We evaluate our method on two real-world mobile traffic datasets. The Shanghai dataset records traffic measurements from 1,042 base stations at 30-minute intervals from August 1 to August 31, 2014, while the Nanjing dataset contains measurements from 1,000 base stations at the same

Table 1. Performance comparison of models for Shanghai and Nanjing

Methods	Shanghai			Nanjing		
	MAE	RMSE	R^2	MAE	RMSE	R^2
STGCN (2018)	0.2382±0.0029	0.3691±0.0041	0.8442±0.0034	0.2702±0.0032	0.4051±0.0048	0.8438±0.0037
DCRNN (2018)	0.2361±0.0030	0.3716±0.0045	0.8421±0.0038	0.2684±0.0027	0.4026±0.0044	0.8457±0.0033
WaveNet(2019)	0.2351±0.0024	0.3672±0.0042	0.8458±0.0035	0.2672±0.0025	0.3981±0.0039	0.8491±0.0029
timegrad (2021)	0.2427±0.0027	0.3731±0.0049	0.8409±0.0042	0.2745±0.0033	0.4163±0.0049	0.8350±0.0039
DiffSTG (2023)	0.2408±0.0026	0.3702±0.0046	0.8433±0.0039	0.2756±0.0031	0.4134±0.0051	0.8373±0.0040
timellm (2024)	0.2414±0.0035	0.3716±0.0048	0.8421±0.0040	0.2698±0.0039	0.4043±0.0042	0.8444±0.0032
MoGERNN (2025)	0.2355±0.0024	0.3667±0.0042	0.8462±0.0035	0.2654±0.0034	0.3972±0.0048	0.8497±0.0036
TALON (2025)	0.2348±0.0023	0.3662±0.0037	0.8466±0.0031	0.2661±0.0037	0.3966±0.0045	0.8502±0.0033
MoE-LLM (ours)	0.2226±0.0021	0.3471±0.0035	0.8623±0.0027	0.2511±0.0028	0.3663±0.0034	0.8722±0.0024
Improvement (%)	5.2%	5.2%	1.9%	5.4%	7.6%	2.6%

temporal resolution from February 2 to March 31, 2021. For both datasets, we construct prediction samples using the first 12 steps as historical observations and the final step as the prediction target.

5.2 Baselines.

We compare MoE-LLM with state-of-the-art baselines from four representative categories. Deep learning-based methods include STGCN [26], DCRNN [10], and Graph WaveNet [22]. MoE-based methods include MoGERNN [29], which employs multiple experts for traffic prediction at un-observed locations. Diffusion-based methods include DiffSTG [21] and TimeGrad [16], while LLM-based method TimeLLM [7] and agent-based method TALON [18]. All baselines are evaluated using their default configurations.

5.3 Implementation Details and Evaluation.

All experiments are conducted on a single NVIDIA RTX 4090 (24GB). We adopt GPT-4.1-nano as the backbone LLM via OpenAI API, with temperature=0.0. We evaluate their performance using three metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2).

5.4 Overall Performance

Table 1 shows that MoE-LLM consistently outperforms all baselines in traffic prediction. In Shanghai, MoE-LLM reduces MAE by 5.2% and RMSE by 5.3% over best baseline; in Nanjing, the improvements are 5.0% and 5.4%, respectively. The performance reveals that existing GNN and diffusion models rely on fixed architectures, while LLM-based methods lack expert specialization. MoGERNN offers learnable routing but relies solely on numerical patterns without semantic reasoning. In contrast, our framework addresses these gaps through two complementary mechanisms. Specifically, the MoE architecture with specialized experts reduces RMSE by avoiding large errors during unexpected changes. The LLM dynamically coordinates expert collaboration based on scene semantics, improving MAE by avoiding inappropriate averaging across conflicting patterns. Consistently high R^2 values across two cities demonstrate effective adaptation to heterogeneous traffic regimes. These designs enable MoE-LLM to achieve consistent performance gains across datasets.

5.5 Ablation Study

We evaluate the contribution of each core component by removing spatial experts, removing temporal experts, and replacing the LLM coordination agent with an MLP gating network or simple

Table 2. Ablation study. Performance comparison of models with different components removed

Variants	Shanghai			Nanjing		
	MAE	RMSE	R^2	MAE	RMSE	R^2
w/o spatial experts	0.3112	0.4566	0.7616	0.3572	0.5374	0.7250
w/o temporal experts	0.2362	0.3681	0.8451	0.2675	0.4024	0.8458
w/o coordination (simple average)	0.2549	0.3867	0.8290	0.2898	0.4413	0.8146
w/o coordination (MLP gating)	0.2314	0.3615	0.8506	0.2668	0.3956	0.8510
MoE-LLM	0.2226	0.3468	0.8625	0.2520	0.3753	0.8659

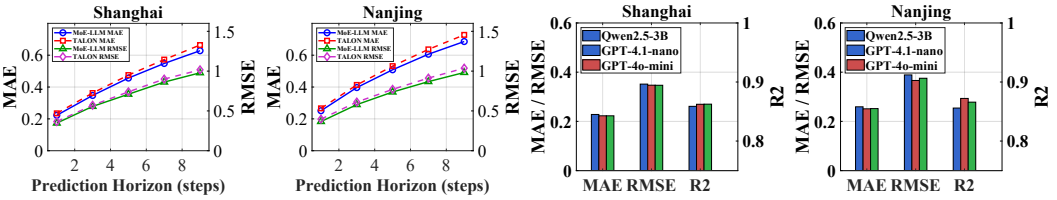


Fig. 3. Multi-step forecasting performance and LLM backbone analysis on the Shanghai and Nanjing datasets. From left to right: multi-step forecasting on Shanghai and Nanjing, and LLM backbone analysis on Shanghai and Nanjing.

averaging. As shown in Table 2, removing spatial experts causes the largest performance degradation, highlighting the importance of cross-base-station dependencies. Removing temporal experts also reduces performance, confirming the value of modeling temporal dynamics. Replacing the LLM agent with MLP gating or simple averaging further degrades accuracy, with simple averaging performing worst. These results demonstrate that MoE-LLM benefits from both complementary expert specialization and LLM-based adaptive coordination.

5.6 Prediction Horizon Analysis

We evaluate multi-step forecasting performance over horizons of 1, 3, 5, 7, and 9 steps. Figure 3 shows that MoE-LLM consistently outperforms TALON across all horizons on both datasets, indicating that MoE-LLM is more effective at mitigating error accumulation over long-range forecasts. This advantage stems from two complementary designs. Multiple experts capture distinct temporal patterns for accurate long-term evolution modeling, while the coordination agent dynamically adjusts expert collaboration to maintain stability over extended horizons.

5.7 LLM Backbone Analysis

We compare three LLM backbones: GPT-4o-mini, GPT-4.1-nano, and Qwen2.5-3B. Figure 3 shows that GPT-4o-mini and GPT-4.1-nano achieve comparable performance, while Qwen2.5-3B yields slightly higher errors. Even the lightweight Qwen2.5-3B consistently outperforms the MLP gating baseline (w/o coordination agent in Table 2), confirming that coordination provides unique advantages such as semantic understanding of spatiotemporal context. The results demonstrate that coordination benefits from greater reasoning capability and greater reasoning capability. Considering the balance of performance, cost, and inference speed, we adopt GPT-4.1-nano as the default backbone.

5.8 Computational Complexity Analysis

We analyze the computational cost of LLM-based coordination using GPT-4.1-nano as the default backbone. On average, each base station consumes approximately 154K tokens and takes about 12.1

Example of Expert Coordination Agent Reasoning: Prediction and Reflection

🔗 **Prediction phase (Station 1):** The timestamp is 2021-03-15 08:00:00, POI distribution: Office 0.6, Retail 0.3, Residential 0.1. Historical traffic sequence shows rising trend over 12 steps (0.32→0.54, mean=0.46). Expert profiles: Geography Expert (reliance=0.35), Function Expert (reliance=0.25), Pattern Expert (reliance=0.20), Temporal Expert (reliance=0.20). Agent Decision: {"Predictions": [0.52], "reason": "Office-dense POI + Monday 8:00 AM indicates commuting peak, supporting higher traffic. Geography Expert and Pattern Expert are most reliable during morning rush hour because they capture spatial congestion propagation. Function Expert is less reliable due to mixed land use in this area ..."} 🔄 **Prediction phase (Station 2):** ...

🔗 **Reflection phase (Station 1):** Previous prediction errors: Geography Expert error=0.0004, Pattern Expert error=0.0025, Function Expert error=0.0081, decomposition models error=0.0093. Agent Decision: {"Geography Expert new reliance": 0.42, "Pattern Expert new reliance": 0.30, "Function Expert new reliance": 0.16, "decomposition models new reliance": 0.12, "reason": "Geography Expert performs best this round, so its reliance increases. Pattern Expert also shows strong performance. Function Expert and decomposition models underperform due to poor handling of sudden spatial fluctuations in office-dominated morning rush scenarios. Decomposition models show highest combined error, so score reduces. ..."} 🔄 **Reflection phase (Station 2):** ...

Fig. 4. Expert coordination agent reasoning example for prediction and reflection.

seconds for inference. These results demonstrate that MoE-LLM is practically viable for large-scale traffic prediction tasks, with per-base-station overhead well within acceptable ranges for real-world deployment scenarios.

5.9 Interpretability Analysis

Beyond quantitative improvements, the experts coordination agent provides interpretable reasoning for both prediction and expert adaptation. As shown in Figure 4, the agent explains how spatio-temporal context, historical traffic patterns, and expert reliability influence the final prediction. For example, in an office-dominated region during morning rush hour, it assigns greater importance to experts that better capture spatial propagation and similar temporal patterns. After the ground truth becomes available, the agent further explains reliance updates according to expert prediction errors, increasing the influence of reliable experts while reducing that of underperforming ones.

6 Conclusion

In this paper, we propose MoE-LLM, a framework that integrates specialized forecasting experts with an LLM-powered coordinator. The LLM dynamically combines temporal and spatial experts based on historical data, POIs, and reliance scores, while a closed-loop feedback mechanism refines expert coordination via prediction error analysis. Future work will extend the single-agent coordinator to a multi-agent framework, where agents specialize in distinct traffic regimes, such as peak and off-peak periods, and negotiate expert contributions via agentic communication.

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